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24-HOUR CLOCK ON DISCRETE LOGIC

IC 555 Timebase with a CD4026 Counter–Display Chain

Submitted by

THE ASTABLE MINDS

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1. Abstract

This project implements a time-of-day clock covering 00:00:00 to 23:59:59 entirely in hardware, with no processor, firmware, or clock module anywhere in the design. The second-by-second reference originates in an NE555 running as an astable multivibrator, and six cascaded CD4026 devices — decade counters that decode straight to seven-segment format on-chip — turn that pulse train into six visible digits on common-cathode displays.

The technical heart of the project is rollover. A CD4026 exposes no binary count, so the wrap points at 60 s, 60 min and 24 h cannot be found by comparing BCD values. They are found instead by pattern matching the segment drive lines: the digits involved in each wrap illuminate segment combinations that occur for no other digit, and a compact NOT/AND network converts those combinations into asynchronous reset pulses.

The circuit was realised on a breadboard for under ₹500 and verified end-to-end by our team. Along the way we ran into a genuine rollover bug of our own — detailed in Section 13 — which we traced and fixed through signal-integrity work on the reset lines and a disciplined, module-by-module build order.

Keywords: *discrete-logic clock, 555 astable, CD4026, common-cathode display, segment decoding, asynchronous reset*

2. Introduction

When timekeeping is delegated to a microcontroller, the entire mechanism collapses into a few opaque lines of code. Building the same function from separate logic packages restores visibility: each conceptual step of the clock — oscillate, count, decode, reset — occupies its own identifiable silicon and its own probeable wires.

Here the oscillation is an NE555 timer, the counting is a chain of six decade counters, and the reset decisions belong to a few gates whose inputs and outputs can be measured live. The transition from 09:59:59 to 10:00:00 is not an abstraction; it is a specific gate output going high on a specific wire at a specific instant.

Choosing the CD4026 shaped the whole project. Where classical designs devote one counter plus one CD4511 decoder to each digit, the CD4026 merges the pair, halving the parts count — but it withholds the binary count value in doing so. Rollover detection therefore cannot follow the textbook, and the workaround — reading the segment lines as a code — is the project's main design contribution.

Digital clocks also make a convenient teaching vehicle precisely because their correctness is so easy to check by eye: a wrong wrap point, a stuck digit, or a missed carry shows up immediately on the display rather than needing a debugger or a serial monitor to uncover. That immediacy is largely what a microcontroller build gives up — the logic is still there, but it is compiled away into an opaque binary, and a fault reveals itself only as a symptom on the display, several layers removed from its cause.

3. Problem Statement

The task: build a 24-hour clock (00:00:00 – 23:59:59) obeying the following constraints.

• Discrete ICs exclusively; microcontrollers, programmed parts and prebuilt modules are barred.

• An IC 555 must generate the timebase on the board itself.

• CD4026 counters must perform the counting and drive the seven-segment displays directly. • Correct wrapping is required: 60 for seconds and minutes, 24 for hours.

• Operation from one +5 V supply, demonstrated on a breadboard.

The defining difficulty follows from the counter choice: with no BCD pins on the CD4026, wrap values must be recognised from the display drive signals alone — the binary-comparison approach taught in most textbooks simply does not apply.

4. Objectives

1. Configure a 555 as an astable oscillator near 1 Hz and confirm the frequency analytically and experimentally.
2. Assemble a six-stage CD4026 cascade organised as seconds, minutes and hours pairs, each pair driving two displays.
3. Work out segment-line logic that fires uniquely on the digit 6 and on the two-digit value 24. 4. Deliver glitch-tolerant asynchronous resets to each pair through NOT/AND networks.
5. Build, fault-find and validate the complete unit on a breadboard, logging each defect and remedy.
6. Document the design at a depth reproducible by second-year ECE students.

5. Components Used

The bill of materials contains only ordinary through-hole stock, totalling about ₹400–500.

LM555 timer	1	Astable oscillator — the clock's 1 Hz heartbeat.
CD4026BE counter	6	One per digit; counts by ten and decodes to seven-segments internally.
Common-cathode 7-segment display	6	Digit readout. Common cathode is required since CD4026 segment outputs drive HIGH.
Hex inverter (74HC04 / CD4069)	1	Provides the inverted terms used by the wrap decoders.
Quad 2-input AND (74HC08 / CD4081)	1	Forms the seconds and minutes reset expressions.

Triple 3-input AND (74HC11 / CD4073)	1	Forms the three-term hours-reset expression detecting 24.
Resistor, 10 kΩ	1	Timing resistor R1 of the 555.
Resistor, 68 kΩ	1	Timing resistor R2 of the 555.
Capacitor, 10 µF electrolytic	1	Timing capacitor setting the ~1 s period with R1 and R2.
Capacitor, 10 nF ceramic	1	Decouples 555 pin 5 for frequency stability.
Resistor, 330 Ω	42	Series element on each segment line; sets LED current and protects logic levels.
Resistor, 220 Ω	—	Added on the reset-logic lines to clean up the reset transition (Section 13).
Capacitor, 0.1 µF ceramic	9	Local supply decoupling at each IC.
Push buttons with 10 kΩ pull-downs	2	Time adjustment via manual clock pulses to MIN-1 and HOUR-1.
Breadboards, +5 V source, jumper wire	—	Physical assembly.

5.1 Why This Counter

Twelve chips (six counters, six CD4511 decoders) would do the same display job the conventional way. Six CD4026s do it alone. The economy is real, but so is the consequence: no binary count is ever available, and Section 8 shows the decoding discipline that replaces it.

6. Block Diagram

Viewed as a system, the clock is a frequency-division pipeline in which each stage passes a slower pulse onward.

- The 555 clocks SEC-1 once per second.
- SEC-1's carry advances SEC-10; a decoder clears both the instant SEC-10 would show 6, so the pair displays 00–59.
- That clearing pulse simultaneously clocks the minutes stage; the minutes pair is bounded to 00–59 by an identical decoder.
- The minutes reset clocks HOUR-1; a three-input decoder clears both hour chips upon 24, bounding them to 00–23.

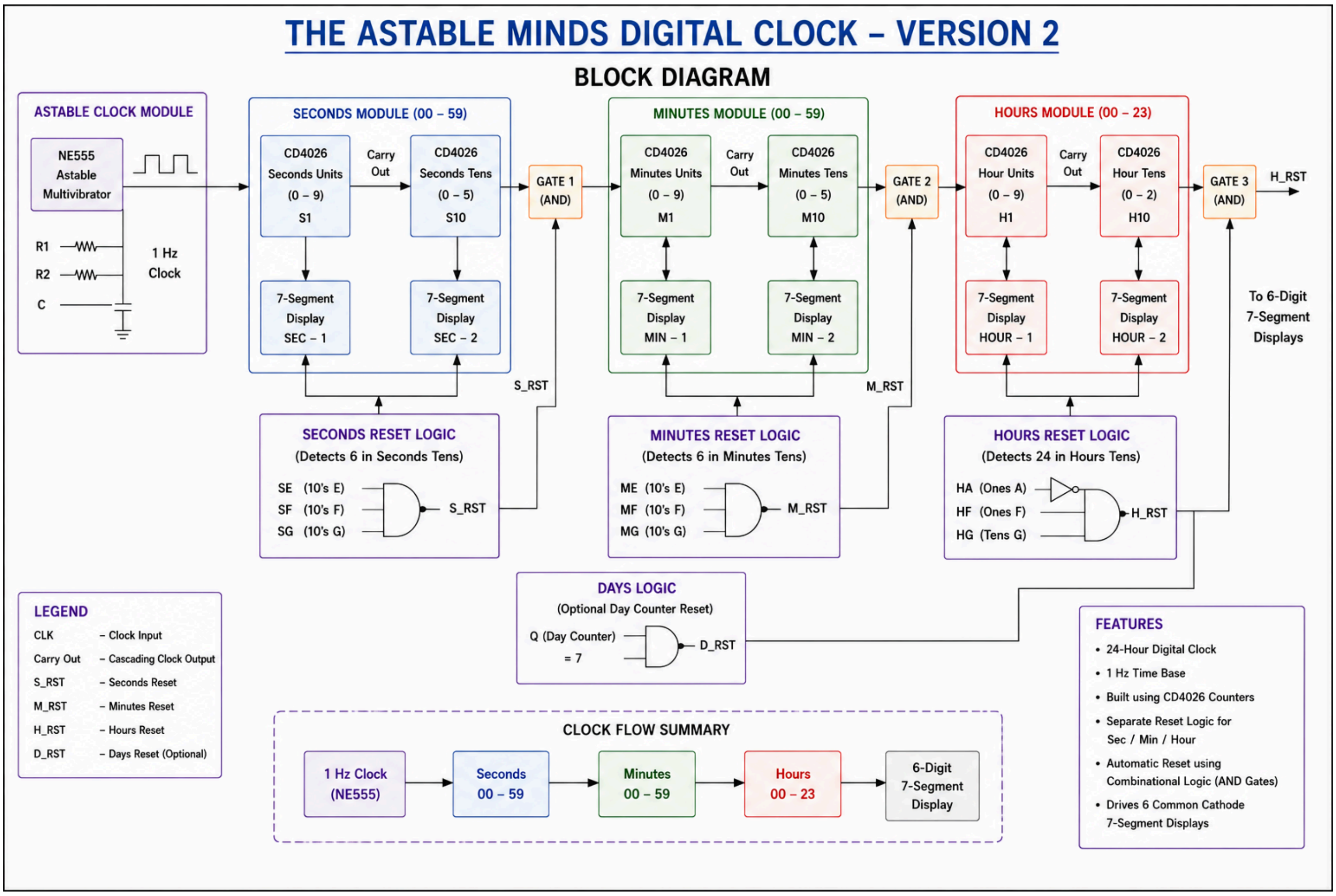


Figure 1: System block diagram — THE ASTABLE MINDS Digital Clock, Version 2.

7. Circuit Diagram

The full schematic realises the block diagram as three cascaded CD4026 pairs — seconds, minutes and hours — each pair followed by its own segment-decode reset network, shown bottom-left of the sheet.

Reading the sheet left to right: the Astable Clock Module in the top-left corner is the entire timebase — a single LM555 with its two timing resistors and capacitor, producing the 1 Hz square wave that drives everything downstream. That signal enters the Seconds Module first, ripples through its two CD4026 stages, and its own reset decoder feeds both the display and the Minutes Module's clock input; the same pattern repeats from Minutes into Hours. The Reset Logic block in the lower-left groups all three NOT/AND decoders together on one sheet, making it straightforward to trace exactly which segment taps — SB, SE, MB, ME, H1A, H1F, H1G, HG — feed each decision, and to follow S_RST, M_RST and H_RST back to the RESET pin they ultimately drive .

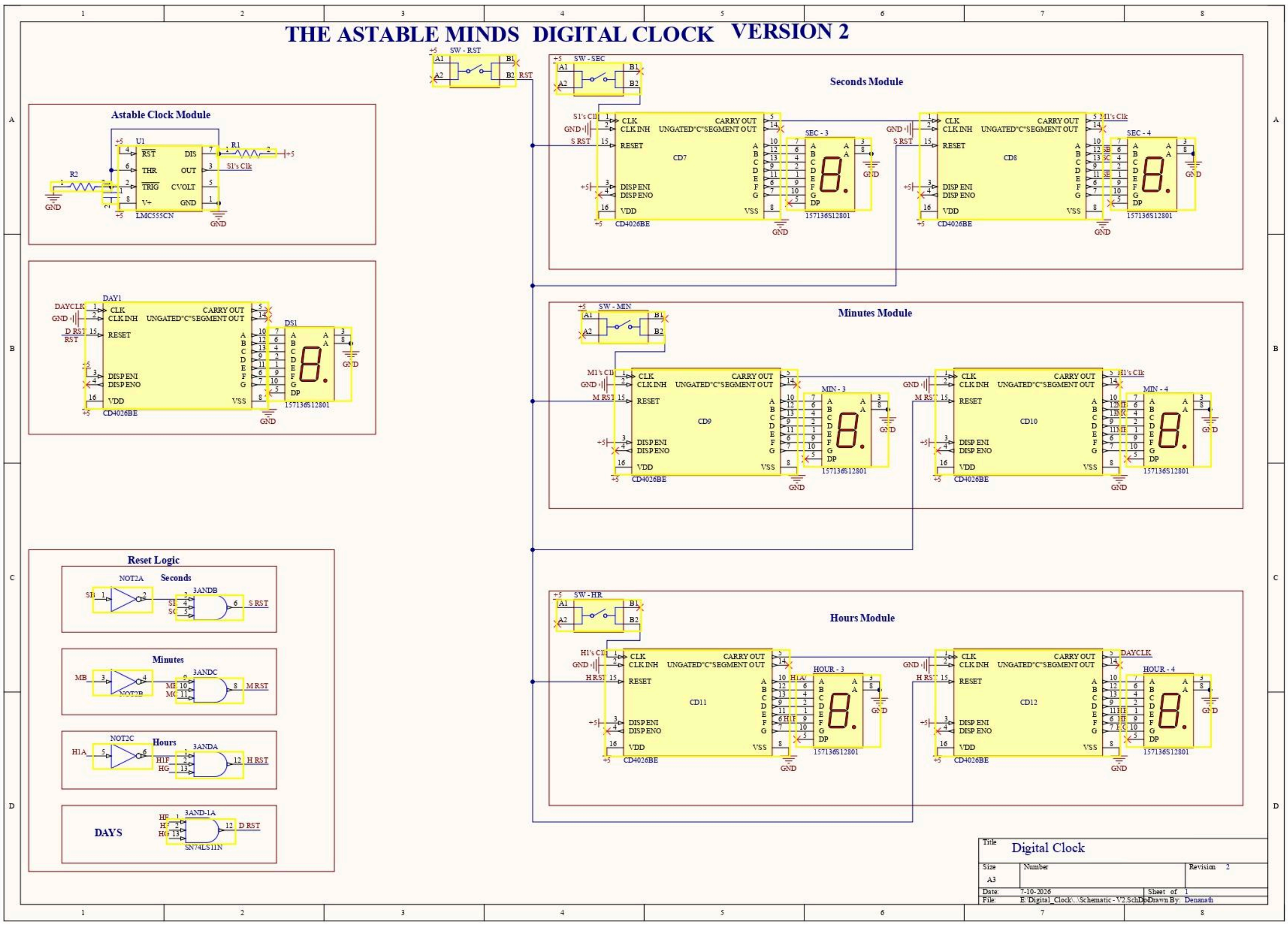


Figure 2: Full circuit schematic, Version 2.

7.1 Per-Chip Wiring Summary (CD4026)

Pin	Name	Destination
16	VDD	+5 V rail, 0.1 µF to ground alongside
8	VSS	Ground rail
1	CLOCK	Preceding carry or reset pulse; 555 output for the first chip
2	CLOCK INHIBIT	Ground (counting is disabled if high or floating)
3	DISPLAY ENABLE IN	+5 V to keep segments active
15	RESET	The group's decode output; clears on HIGH
5	CARRY OUT	Next stage's CLOCK — one pulse every ten counts

Segment outputs a–g (pins 10, 12, 13, 9, 11, 6, 7) go to the display pins, each through a 330 Ω resistor; display commons go to ground (common cathode). The decode pick-off points attach on the counter side of each 330 Ω resistor — Section 13 explains what goes wrong otherwise.

8. Working Principle

8.1 Generating the Second

In astable mode the 555's capacitor charges via R1 + R2 and discharges via R2, so pin 3 toggles indefinitely. Section 9 shows the period lands near one second for the chosen parts. RESET (pin 4) sits at +5 V; pin 5 carries a 10 nF stabilising capacitor.

8.2 The Ripple Chain

A CD4026 advances on each rising clock edge and emits one CARRY OUT pulse per ten counts, which becomes the next digit's clock. Absent any reset logic, each two-chip group would simply count to 99; the decoders are what carve 60 and 24 out of that range.

8.3 Wrap Detection from the Segment Lines

For seconds and minutes the target is a 6 on the tens digit. Segment b is dark only for 5 and 6; within that pair, segment e distinguishes them, being dark on 5 and lit on 6. Therefore:

RESET = (NOT b) · e

is satisfied by 6 and by nothing else. As the tens digit ticks to 6 the expression rises, both counters clear, and the 6 lives for mere hundreds of nanoseconds — invisibly brief, so the display reads 59 then 00. The same edge clocks the next stage.

For hours the target is the two-digit value 24, decoded as a conjunction: the tens digit is 2 (only 0, 1 and 2 ever appear there, and segment g lights solely for the 2, so a single tap identifies it outright); the units digit is 4 (f·g alone would also accept 5, 6, 8 and 9, so an inverted-a term is added to leave only 4):

HRS RESET = HG · $\bar{a}$ · f · g

At the instant the readout would show 24, both hour chips clear and the day restarts at 00.

8.4 Reset Semantics of the CD4026

Pin 15 acts immediately and without a clock: a HIGH forces the count to zero and pins it there until released, after which the next rising clock edge counts 0 → 1. The wrap pulses here are self-terminating — the clear removes the pattern that produced it — yielding very narrow pulses that CMOS logic normally resolves cleanly. Any RESET pin that has no logic drives must be tied to ground, because a floating CMOS input will pick up noise and clear the counter at random.

9. Design Calculations

9.1 Astable Frequency

For R1 = 10 kΩ, R2 = 68 kΩ, C = 10 μF, the standard relation gives:

$f = 1.44 / ((R1 + 2R2)C) \approx 0.986$ Hz, $T \approx 1.014$ s

— about 1.4% away from a true second, before component tolerances even enter.

9.2 Mark–Space Ratio

Duty = (R1 + R2) / (R1 + 2R2) ≈ 53.4%. This is irrelevant to accuracy, since only rising edges clock the counters.

9.3 LED Current

Each lit segment draws about (5 V − 2 V − output drop) / 330 Ω ≈ 7–8 mA — adequately bright and safely inside the CD4026's drive capability.

9.4 Drift

RC timing wanders with temperature and capacitor ageing; minutes-per-day error is the honest expectation and is acceptable for a demonstrator.

Assembly followed a strict prove-then-extend order, so each newly observed fault could be attributed to the most recent addition only.

7. Rails and decoupling: +5 V and ground distributed, mid-rail gaps bridged, 0.1 µF beside every package.
8. Oscillator alone: R1, R2 and C wired around the 555; the ~1 Hz square wave verified before any counter joined.
9. One digit: SEC-1 tested through a full 0–9 cycle on its display.
10. Seconds group: SEC-10 added on the carry; the ·e decoder installed; 00–59–00

confirmed. b̄ 11. Minutes group: cloned from seconds, clocked by the seconds reset line.

12. Hours group: the three-input AND wired; the wrap at 24 checked by manual clocking.
13. Setting controls: two push buttons (with pull-downs and debounce capacitors) pulse the MIN-1 and HOUR-1 clocks for manual adjustment.

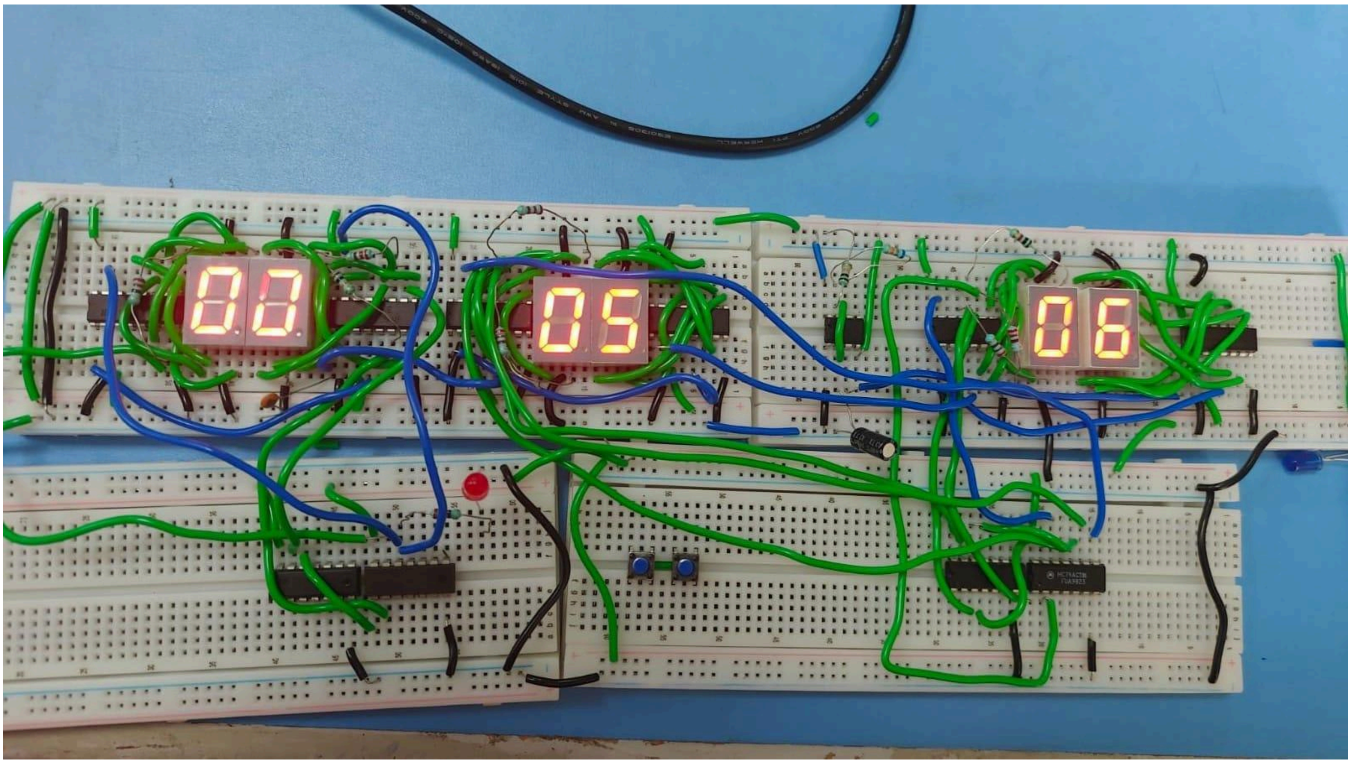
11. Testing and Validation

1 Hz reference	60 ticks timed against a stopwatch	≈ 0.99 Hz, matching Section 9
Per-digit count	Every display watched over a full decade	Correct on all six
60 s wrap	Observed at the minute boundary	59 → 00, carry delivered, transient 6 invisible
60 min wrap	Driven quickly with the set button	59 → 00 with carry into hours
24 h wrap	Preset to 23:59 and awaited	23:59:59 → 00:00:00 achieved
Decode levels	Meter on the tap points	Clean 0 / ~4.5 V swing after the 220 Ω fix (Section 13)
Pin-15 behaviour	Reset asserted manually	Immediate clear and hold, per datasheet

12. Results

The clock cycles continuously through the full 24-hour range of six digits, with every wrap mechanism dependable in repeated trials. Untrimmed timebase error stayed below 2%, in line with the RC oscillator norm.

Nine ICs in total; supply current 150–250 mA at 5 V depending on the digits lit; component spend under ₹500.



13. Challenges Faced

The build was not without its own genuine debugging story, kept here in detail because the diagnosis is where the learning lives.

13.1 The Seconds/Minutes Wraparound Bug

Symptom: our seconds and minutes pairs counted 00 through 59 correctly, but instead of wrapping cleanly back to 00 at the boundary, the display rolled on into 60, 61 and beyond before eventually resetting — rather than the clean 59 → 00 transition the design called for.

Diagnosis: the reset pulse generated by our NOT/AND decode network was reaching pin 15, but the edge into the reset pin was not clean or fast enough over the breadboard's longer signal traces to reliably force a clear right on the boundary. As a result, the counter would occasionally ripple one or two extra counts before the reset caught up — exactly the kind of signal-integrity issue that is easy to miss when only the logic equation, and not the physical wiring, is checked.

Fix 1 — 220 Ω resistors on the reset lines: adding a 220 Ω resistor at each reset-logic pin cleaned up the signal feeding into pin 15, giving a sharper, more reliable transition and eliminating the extra counts. This brought the rollover back to a clean 59 → 00 in every trial.

Fix 2 — module-by-module debugging: rather than wiring the seconds, minutes and hours stages together and hunting for the fault across the whole cascade, we isolated and fully validated each module on its own — seconds first, then minutes, then hours — before cascading them together. This made the wraparound bug straightforward to localise, and became our standard approach for the rest of the build.

Lesson: on a reset network built from discrete gates, correctness on paper is not the same as correctness on the breadboard — the physical reset edge has to be verified with a meter or scope, not assumed from the logic equation alone. Debugging one self-contained module at a time, rather than the full cascade, made every subsequent fault far faster to locate.

14. Applications

- Laboratory teaching rig: oscillation, cascading, decoding and reset are all live, probeable signals.
- Budget clock for settings tolerant of small daily drift.
- Reusable counter core — relocating the decode points converts it into a countdown timer, event tally, or production-cycle counter.
- Scheduled switching (bells, pumps) by decoding selected times into control outputs.

15. Future Scope

- Swap the 555 for a 32.768 kHz crystal and CD4060 divider — drift improves from minutes to seconds per day; the most consequential upgrade available.
- Battery hold-up so the count rides through mains interruptions.
- Alarm feature: decode a target time and switch a buzzer through a transistor.
- 12-hour readout with an AM/PM indicator via extra decoding.
- Soldered PCB in place of the breadboard for durability and presentation.
- A parallel ESP32/NTP implementation for a cost–accuracy–complexity comparison against the discrete build.

16. Conclusion

Nine discrete ICs — a 555, six CD4026 counters and three gate packages — were shaped into a validated 24-hour clock satisfying every constraint: local timebase, direct display drive, and correct wraps at 60 seconds, 60 minutes and 24 hours.

What sets the design apart is where the wraps are detected: on the seven-segment lines, a necessity imposed by the CD4026's missing BCD outputs. Producing decodes that are both minimal and provably unique — $\cdot e$ for the 6, g plus $\bar{a} \cdot f \cdot g$ for the 24 — meant disciplined work with the segment truth table. $\bar{b}$ Getting the build itself to behave, including tracking down and fixing our own wraparound bug with 220 Ω resistors and a module-by-module rebuild strategy, was just as important a part of the learning as the logic design.

The result is an affordable, repeatable and instructive project, built and debugged end-to-end by our own team, squarely at second-year ECE level.

17. References

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